UNIVERSITY OF GUJRAT
A WORLD CLASS UNIVERSITY

Department of IT

COURSE DESCRIPTION

Course Code	MATH-314
Course Title	Linear Algebra
Credit Hours	03
Category	Computing Supporting Course
Prerequisite	MATH-107, MATH-110
Amis and Objectives	Introduces the foundations of linear algebra as they apply to Computer Science, focusing on providing a solid theoretical foundation for further work. Further, this course aims to develop understanding and appreciation of the finite nature inherent in most Computer Science problems and structures through study of matrices and determinants, systems of linear equations, vectors and vector spaces. In this course more emphasis shall be given to statistical and probabilistic formulation with respect to computing aspects.
Learning Outcomes	One can use the matrices and equations to form programs and games in Computer Science.
Course Outline / Syllabus	Vectors, Vector spaces, Matrices & Determinants, Cofactor and Inverse, Rank, Linear independence, Solution of system of linear systems, Positive Definite matrix, Linear Transformations, Operations on matrices, Inner products, Orthogonality and least squares, Eigen value & Eigenvectors, Applications to systems of equations and to Geometry, Singular Value Decomposition.
Text Book	Howard Anton, Chris Rorres, Elementary Linear Algebra: Applications Version, 10th edition, Wiley.
Reference Material	1. Bernard Kolman, David Hill, Elementary Linear Algebra with Applications, 9th edition, Prentice Hall PTR, 2007. 2. David C. Lay, Linear Algebra and Its Applications, 2nd edition, 2000.
Assessment Criteria	5% weightage of Quiz, 10 % weightage of Assignments and 10% weightage of Presentation. 25 % weightage of Mid Term Exam and 50 % weightage of Final Exam.

Grading System

Marks in Percentage	***Letter Grade***	***Numeric value of Grade***	***Description***
85+	A+	4.00	Exceptional
80-84	A	3.70	Outstanding
75-79	B+	3.40	Excellent
70-74	B	3.00	Very Good
65-69	B-	2.50	Good
60-64	C+	2.00	Average
55-59	C	1.50	Satisfactory
50-54	D	1.00	Pass
49 and below	F	0.0	Fail
	W		Withdrawal
	I		Incomplete

Week	**Lecture**	**Topic**	**Source**	**Recommendations for**

			(Book-Chapter No. Section No.)	Learning Activities (Mention Assignments, Test, Quizzes, Practical, Case Study, Projects, Lab Work or Reading Assignments)
1	1	• Matrices and Matrix Operation	Howard Anton, Chris Rorres, Elementary Linear Algebra: Applications Version, 10th edition. Chapter # 01, Sec 1.3	
	2	• Matrices and Matrix Operation	Chapter # 01,Exercise 1.3	
2	3	• Inverses, Algebraic properties of matrices	Chapter # 01, Sec 1.4	
	4	• Inverses, Algebraic properties of matrices	Chapter # 01, Exercise 1.4	
3	5	• Elementary matrices	Chapter # 01, Sec 1.5, Exercise 1.5	
	6	• A method for finding inverse	Chapter # 01, Sec 1.5, Exercise 1.5	
4	7	• Quiz		Subjective type quiz
	8	• More on linear systems and invertible matrices	Chapter # 01, Sec 1.6, Exercise 1.6	
5	9	• Diagonal, Triangular and Symmetric matrices	Chapter # 01, Sec 1.7, Exercise 1.7	
	10	• Systems of Linear Equations	Chapter # 01, Sec 1.1, Exercise 1.1	
6	11	• Gaussian Elimination Method	Chapter # 01, Sec 1.2, Exercise 1.2	Assignment will be given
	12	• Gauss Jordan Elimination Method	Chapter # 01, Sec 1.2, Exercise 1.2	
7	13	• Quiz		Subjective type Quiz
	14	• Determinants	Chapter # 02, Sec 2.1	

8	15	• Determinants by Cofactor Expansion	Chapter # 02, Sec 2.1, Exercise 2.1	
	16	• Determinants by Row Reduction	Chapter # 02, Sec 2.2, Exercise 2.2	
9	17	• Mid Term Exam		Objective and Subjective
	18	Mid Term Exam		
10	19	• Properties of Determinants	Chapter # 02, Sec 2.3	
	20	Cramer's Rule	Chapter # 03, Sec 2.3, Exercise 2.3	
11	21	• Quiz		Subjective type Quiz
	22	• Vectors in 2-space, 3-space and n-space	Chapter # 03, Sec 3.1, Exercise 3.1	
12	23	• Norm, Dot Product	Chapter # 03, Sec 3.2, Exercise 3.2	
	24	• Orthogonality	Chapter # 03, Sec 3.3, Exercise 3.3	
13	25	• Cross Product	Chapter # 03, Sec 3.5, Exercise 3.5	
	26	• Real vector Spaces	Chapter # 04, Sec 4.1, Exercise 4.1	
14	27	• Subspaces	Chapter # 04, Sec 4.2, Exercise 4.2	
	28	• Eigenvalues and Eigenvectors		
15	29	• Inner Products	Chapter # 06, Sec 6.1, Exercise 6.1	
	30	• Angle and Orthogonality in Inner Product Spaces	Chapter # 06, Sec 6.2	
16	31	• Problems		
	32	• Revision of Course		